THOBEKA THOBILE BUSISIWE NKOSI

SOFTWARE ENGINEERING | JAVA & BACKEND DEVELOPMENT | DATA ENGINEERING

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PROFESSIONAL SUMMARY

Software Engineering student at WeThinkCode_ (NQF 6; completing December 2026) with a BSc in Life Sciences and practical experience in Java, object-oriented design, concurrent client-server applications, REST APIs and automated testing. Built Java systems using sockets, threads and JUnit, and delivered a deployed data platform using Python, PostgreSQL, dbt and Apache Airflow. Analytical, collaborative and motivated to build reliable solutions in a fast-paced engineering environment.

TECHNICAL SKILLS

Languages: Java 21, Python, SQL, TypeScript, JavaScript, Bash

Java & backend: OOP, concurrency and multithreading, Java Sockets, JUnit 5, Maven, Jackson/JSON, REST APIs, microservices, Apache ActiveMQ (JMS)

Data engineering: ETL/ELT, data quality, pandas, PyArrow/Parquet, PostgreSQL, Apache Airflow, dbt, DuckDB

DevOps & web: Git, GitHub Actions, CI/CD, Docker, Kubernetes, Linux, FastAPI, React, Flask

EDUCATION

WeThinkCode_ | Occupational Certificate in Software Engineering (NQF 6) • Sep 2025 – Dec 2026 (expected)
Data Engineering elective: Python data pipelines, extraction and transformation, SQL, data validation, orchestration and analytical modelling. Core work includes Java OOP, distributed applications, testing and CI/CD.

Sefako Makgatho Health Sciences University | BSc Life Sciences (NQF 7) • Completed 2025

Majors: Biochemistry and Physiology. Strong foundation in scientific analysis, research and evidence-based problem solving.

Midlands Community College | National Senior Certificate (Upgraded Matric) • Completed 2019

Phendukani Full Service High School | National Senior Certificate (Matric) • Completed 2018

SELECTED TECHNICAL PROJECTS

Robot World | Concurrent Multiplayer Java Application | GitHub

Java 21 • Java Sockets • Multithreading • Maven • JUnit 5 • Jackson

- Co-developed a multiplayer robot simulation using socket-based client-server communication and JSON command/response messages.
- Implemented multithreaded server operations using Java sockets to support concurrent client connections, enabling simultaneous robot interactions in a shared world.

HealthSafe | Java Microservices & Messaging | GitHub

Java • REST APIs • Apache ActiveMQ/JMS • Docker • Kubernetes

- Worked on a hospital operations system integrating Java microservices for ward management, staffing, alert levels and equipment failures.
- Connected services through REST and asynchronous ActiveMQ queues/topics; processed legacy CSV records and supported containerized execution.

SELECTED TECHNICAL PROJECTS (CONTINUED)

AirAtlas | Deployed Environmental Data Platform | GitHub · Live App

Python • PostgreSQL • Apache Airflow • dbt • pandas • Parquet

- Built an end-to-end pipeline for South African PM2.5 and PM10 readings, including API ingestion, raw-data retention, validation, deduplication and weather enrichment.
- Loaded a PostgreSQL warehouse, built tested dbt analytics models, orchestrated workflow stages with Airflow and deployed an interactive dashboard.

PulseGrid AI | Emergency Operations Dashboard | GitHub

Python • FastAPI • React • WebSockets • Leaflet

- Built a simulated emergency operations dashboard with patient triage, hospital capacity monitoring and routing based on priority and queue pressure.
- Connected FastAPI services to a React interface and used WebSocket updates to visualize changes after simulated patient intake.

Pinky-Pact | Blockchain Accountability MVP | GitHub · Live App

TypeScript • React • Solana Devnet • Anchor/Rust • AI integrations

- Co-developed a habit-accountability application with wallet connection and an on-chain pact-creation transaction on Solana devnet.
- Contributed to a responsive demo with AI coaching and witness/settlement prototype flows; real on-chain pact creation is implemented.

BeyondPink | Brea | Inclusive Health-Education Experience | GitHub · Live App

React • TypeScript • Leaflet • Interactive learning • AI assistant

- Created a responsive breast-health awareness site and interactive learning prototype with educational content in five South African languages.
- Built guided learning and support experiences, an interactive care-directory map and an AI educational assistant with optional voice controls.

EXPERIENCE & LEADERSHIP

Work Readiness Facilitator (IDC role) | WeThinkCode_ Oct 2025 – Present

Facilitate workshops and mentor learners on workplace communication, interview preparation, CV writing and professional presentations.

Administrative Assistant (part-time) | Dr SS Nkosi Dental Surgery Jun – Dec 2020

Supported patient scheduling, records, data entry, billing and daily clinic operations.

CERTIFICATIONS & RECOGNITION

Certificates: Kubernetes and Cloud Native Associate (KCNA), The Linux Foundation (2026); Kubernetes and Cloud Native Essentials (2026); Scientific Computing with Python, freeCodeCamp (2025); GenAI course for Software Engineers (2025).

Recognition: 3rd Place, Monkey & River x Momentum Hackathon (2026); 3rd Place, Cisco WISE Hackathon (2026).